Projective Schemes and Gluing: Building Projective Space from Affine Windows

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§1.1 The projective line is assembled from affine pieces

Earlier projective geometry used homogeneous coordinates to add points at infinity and remove affine exceptions. Scheme theory adds another layer: every affine window comes with a ring of functions, and the windows must be glued compatibly.

The construction that does this is Proj. It is to graded rings what Spec is to ordinary rings, but the grading forces an essential degree-zero condition on every chart. We follow the construction of Proj and twisting sheaves in the *Stacks Project*.[1]

The best first example is $\mathbb{P}^1$. Take two affine lines:

$$U_0 = \operatorname{Spec} R[t], \qquad U_1 = \operatorname{Spec} R[s].$$

On the region where both coordinates are nonzero, glue them by

$$s = \frac{1}{t}.$$

The two copies overlap on $\operatorname{Spec} R[t, t^{-1}] \cong \operatorname{Spec} R[s, s^{-1}]$. The transition homomorphism sends $s \mapsto t^{-1}$. In the homogeneous coordinates $[x_0 : x_1]$, the first chart has $t = x_1/x_0$ and the second has $s = x_0/x_1$, which verifies the relation $ts = 1$. The result is the projective line. The point is not the formula alone, but the construction style:

projective space is made by gluing affine charts.

§1.2 Homogeneous primes and the degree-zero chart algebra

Let

$$S = S_0 \oplus S_1 \oplus S_2 \oplus \cdots$$

be a graded ring. Projective geometry only respects homogeneous equations, because

$$[x_0 : \cdots : x_n] = [\lambda x_0 : \cdots : \lambda x_n].$$

The construction $\operatorname{Proj} S$ uses homogeneous prime ideals not containing the irrelevant ideal.

This definition sounds heavier than the picture, but it encodes a familiar rule: equations in projective space must be homogeneous.

The scheme-theoretic projective space is

$$\mathbb{P}^n_R = \operatorname{Proj} R[x_0, \ldots, x_n],$$

where the variables have degree 1.

[1] See https://stacks.math.columbia.edu/tag/01M3, especially the standard-open construction and the subsequent sections on invertible sheaves.

The standard affine open where $x_i \neq 0$ is written

$$D_+(x_i).$$

On it, the ratios

$$\frac{x_j}{x_i}$$

serve as affine coordinates.

Thus $\mathbb{P}^n$ is covered by $n+1$ affine schemes. Projective geometry is global, but it is computed through affine windows.

For $S = R[x_0, \ldots, x_n]$, the ideal

$$S_+ = (x_0, \ldots, x_n)$$

is called the irrelevant ideal. It corresponds to the forbidden point where all homogeneous coordinates vanish at once:

$$[0 : \cdots : 0],$$

which is not a projective point.

Thus $\operatorname{Proj} S$ uses homogeneous prime ideals that do not contain all positive-degree elements. If we included S_+ for a polynomial ring over a field, it would lie outside every standard open $D_+(x_i)$ and would represent the prohibited simultaneous zero of the homogeneous coordinates. Excluding every homogeneous prime containing S_+ ensures the standard opens cover $\operatorname{Proj} S$. This is the algebraic justification for the nonzero-coordinate rule.

For a homogeneous element $f \in S$, the standard open $D_+(f)$ is affine. Its coordinate ring is the degree-zero part of the localization:

$$\mathcal{O}(D_+(f)) = (S_f)_0.$$

This formula explains why ratios appear in projective coordinates. Localizing at x_i allows division by powers of x_i, and taking degree zero keeps only scale-invariant expressions such as

$$\frac{x_j}{x_i}.$$

Indeed, for $S = k[x_0, \ldots, x_n]$, the degree-zero localization is

$$\begin{aligned}(S_{x_i})_0 &\cong k\left[\frac{x_0}{x_i}, \ldots, \widehat{\frac{x_i}{x_i}}, \ldots, \frac{x_n}{x_i}\right] \\ &\cong k[t_1, \ldots, t_n].\end{aligned}$$

The hat omits $x_i/x_i = 1$. Its spectrum is the affine space $\mathbb{A}^n_k$, so the standard projective chart and its regular functions follow from the degree-zero localization.

§1.3 Three Charts of a Conic

For $\mathbb{P}^2$, the standard opens are

$$D_+(x), \qquad D_+(y), \qquad D_+(z).$$

On $D_+(z)$, use

$$X = \frac{x}{z}, \qquad Y = \frac{y}{z}.$$

On $D_+(x)$, use

$$U = \frac{y}{x}, \qquad V = \frac{z}{x}.$$

Each chart is affine, but the overlap maps must agree on triple intersections. On $D_+(z) \cap D_+(x)$, the coordinates satisfy

$$U = \frac{Y}{X}, \qquad V = \frac{1}{X}, \qquad X = \frac{1}{V}, \qquad Y = \frac{U}{V}.$$

Substitution in either direction gives the identity transition map upon composition; similar ratio identities handle the third chart. These are the actual gluing data defining $\mathbb{P}^2$.

This is the practical meaning of projective schemes: use affine algebra locally, then glue.

Consider the projective conic

$$xz - y^2 = 0.$$

On the chart $z \neq 0$, set $X = x/z$ and $Y = y/z$. The equation becomes

$$X = Y^2.$$

On the chart $x \neq 0$, set $U = y/x$ and $V = z/x$. The equation becomes

$$V = U^2.$$

The same projective curve has multiple affine faces. A single affine chart may hide what happens at infinity, but the projective object keeps all charts at once.

For the conic

$$xz - y^2 = 0,$$

the chart $y \neq 0$ uses coordinates

$$A = \frac{x}{y}, \qquad B = \frac{z}{y}.$$

The equation becomes

$$AB = 1.$$

So in this chart the conic looks like a hyperbola. In another chart it looked like a parabola. This is a concrete reminder that affine appearances depend on the chosen window.

The projective curve is the object obtained by holding all these windows together.

For a single affine hypersurface equation $f(x, y) = 0$, homogenizing to its total degree gives a projective equation

$$F(x, y, z) = 0.$$

For a general affine *ideal*, however, homogenizing a few chosen generators is not by itself a safe definition of scheme-theoretic projective closure. One must homogenize the ideal appropriately (or saturate the resulting homogeneous ideal with respect to the homogenizing coordinate). This distinction matters if embedded or nilpotent structure is to survive correctly.

So Proj is not merely a notation for old projective varieties. It is the projective version of the local-function viewpoint developed in scheme theory.

§1.4 From Functions to Sections

Projective geometry naturally leads to line bundles. The coordinates of $\mathbb{P}^n_k$ scale simultaneously; an ordinary polynomial of degree $d > 0$ is multiplied by λ^d under $[x_0 : \cdots : x_n] \mapsto [\lambda x_0 : \cdots : \lambda x_n]$, so it cannot be a globally defined scalar function on projective space. Instead, the transition factors $(x_i/x_j)^d$ between affine charts define the twisting sheaf $\mathcal{O}(d)$. For $d \geq 0$, its global sections are exactly homogeneous polynomials of degree d in $x_0, \ldots, x_n$. A homogeneous equation of degree d is better thought of as a section of $\mathcal{O}(d)$ than as an ordinary global function.

This explains why projective varieties have fewer global regular functions than affine varieties. Projective geometry shifts attention from functions to sections of line bundles.

This remark points back to divisors and Riemann–Roch: projective geometry, line bundles, and spaces of sections are parts of the same story.

The twisting sheaves therefore complete the passage from homogeneous coordinates to projective schemes. The same graded algebra that builds $\operatorname{Proj} S$ also supplies its affine windows and its line bundles: projective geometry has been reconstructed from homogeneous algebra and gluing while retaining the local functions invisible in a purely incidence-based picture.